

Companion XLIX: The Velocity Power Spectrum in the Relaxation-Driven Cyclic Cosmology

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Abstract

The velocity power spectrum provides a direct probe of the dynamical evolution of large-scale structure. Unlike the density power spectrum, which measures the amplitude of fluctuations, the velocity spectrum measures the rate at which structure grows and the coherence of cosmic flows. In the standard cosmological model, the velocity field is governed by the linear growth rate and the continuity equation. In the Relaxation-Driven Cyclic Cosmology (RDCC), the scale-dependent evolution of the gravitational potential modifies the velocity divergence, the amplitude of bulk flows, and the coupling between baryons and dark matter. This Companion develops a continuous description of the velocity power spectrum in RDCC, deriving how the relaxation scale influences the growth rate, the velocity divergence spectrum, and the observational signatures in redshift-space distortions. The resulting framework provides a distinctive dynamical signature of the RDCC gravitational field.

1 Introduction

The cosmic velocity field encodes the dynamical evolution of large-scale structure. While the density field describes the spatial distribution of matter, the velocity field describes how matter flows through the Universe. These flows are driven by the gravitational potential and provide a direct probe of the growth rate of structure.

In the standard cosmological model, the velocity field is governed by the Euler and continuity equations, and its power spectrum is determined by the linear growth rate. In RDCC, the gravitational potential evolves differently. The relaxation mechanism suppresses the decay of small-scale modes and enhances the coherence of large-scale modes. This modifies the velocity divergence, the amplitude of bulk flows, and the coupling between baryons and dark matter. The velocity power spectrum becomes a direct probe of the relaxation scale k_{rel} .

2 Velocity Divergence in RDCC

The velocity divergence is defined as

$$\theta(\mathbf{k}, a) = \nabla \cdot \mathbf{v}(\mathbf{k}, a). \quad (1)$$

In linear theory, the continuity equation gives

$$\theta(k, a) = -aHf(a)\delta(k, a), \quad (2)$$

where $f(a)$ is the growth rate.

In RDCC, the density contrast evolves as

$$\delta_{\text{RDCC}}(k, a) = \delta(k, a) \left[1 - \chi_\delta \left(\frac{k}{k_{\text{rel}}} \right)^\alpha \right], \quad (3)$$

and the growth rate becomes scale-dependent:

$$f_{\text{RDCC}}(k, a) = f(a) \left[1 - \chi_f \left(\frac{k}{k_{\text{rel}}} \right)^\alpha \right]. \quad (4)$$

The velocity divergence is therefore modified in a scale-dependent manner.

3 The RDCC Velocity Power Spectrum

The velocity divergence power spectrum is defined as

$$P_{\theta\theta}(k) = \langle \theta(\mathbf{k}) \theta^*(\mathbf{k}) \rangle. \quad (5)$$

In RDCC, this becomes

$$P_{\theta\theta, \text{RDCC}}(k) = P_{\theta\theta}(k) \left[1 - \chi_\theta \left(\frac{k}{k_{\text{rel}}} \right)^\alpha \right]. \quad (6)$$

This modification suppresses small-scale velocity power and enhances large-scale coherence, reflecting the relaxation mechanism.

4 Bulk Flows and Coherent Velocities

Bulk flows measure the coherent motion of matter on large scales. They are given by

$$\mathbf{V}(R) = \int d^3k W(kR) \mathbf{v}(\mathbf{k}), \quad (7)$$

where W is a window function.

In RDCC, the enhanced coherence of large-scale modes increases the amplitude of bulk flows:

$$V_{\text{RDCC}}(R) = V(R) \left[1 + \chi_V \left(\frac{k_{\text{rel}}}{k(R)} \right)^\alpha \right]. \quad (8)$$

This produces a distinctive RDCC signature in peculiar velocity surveys.

5 Redshift-Space Distortions

Redshift-space distortions (RSD) arise from the mapping between real-space and redshift-space positions. The RSD power spectrum is given by

$$P_s(k, \mu) = P(k) \left[1 + f\mu^2 \right]^2, \quad (9)$$

where μ is the cosine of the angle between the wave vector and the line of sight.

In RDCC, the growth rate becomes scale-dependent:

$$f_{\text{RDCC}}(k) = f \left[1 - \chi_f \left(\frac{k}{k_{\text{rel}}} \right)^\alpha \right]. \quad (10)$$

The RSD signal therefore acquires a scale-dependent modification that reflects the relaxation scale.

6 Conclusion

The velocity power spectrum in the Relaxation-Driven Cyclic Cosmology reflects the scale-dependent evolution of the gravitational potential. The relaxation mechanism modifies the velocity divergence, the amplitude of bulk flows, and the coupling between baryons and dark matter. These effects produce distinctive signatures in the velocity power spectrum, bulk flow measurements, and redshift-space distortions. The present Companion establishes the theoretical foundation for future studies of cosmic dynamics, peculiar velocities, and the growth of structure in RDCC.

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